

Applied AI Engineer with 4+ years across industry and research. I design and ship production-ready ML systems — from data pipelines to deployed models — with a focus on deep learning, multimodal architectures, and LLM-based applications. I bring a rare combination: hands-on research depth (FAIR/PNRR, computational neuroscience, publications) and real industry delivery (92% accuracy in production gesture recognition, demand forecasting pipelines for retail). Currently building expertise in RAG systems, LLM agents, and LangChain/LangGraph. Available for B2B contracts and freelance collaborations. Remote. EU timezone.
Core stack: Python (PyTorch/Lightning, fastai), CUDA, LangChain · Docker · CUDA · HPC · Git · Linux

SKILLS

Machine Learning	Supervised/unsupervised learning, meta-learning (MAML-style), self-supervised (masked autoencoders), transfer learning, few-shot learning, time-series modeling, Bayesian modeling, hyperparameter optimization, cross-validation, model selection.
Deep Learning	CNNs, transformers, autoencoders/VAEs, GANs, multimodal learning (vision+symbolic/text), sequence models, embeddings, representation learning, computer vision, contrastive learning.
Data/Experimentation	Data acquisition & labeling, feature engineering, large-scale data processing, experiment design, A/B-like benchmarking, statistical testing, error analysis.
MLOps & Systems	Reproducible pipelines, Dockerized environments, Git-based workflows, HPC/cluster training, on-device inference & optimization.
XAI & Analytics	Saliency/attribution, embedding visualization, dimensionality reduction, interpretability reports.
LLM & Agentic AI	RAG systems, LangChain · LangGraph, vector databases (Pinecone), AI agents, function calling, LLMs embeddings, LlamaIndex.

TECHNICAL EXPERIENCE

AI Researcher - GROUNDEEP Project (FAIR – Future Artificial Intelligence Research, PNRR) Spoke 10: Bio-socio-cognitive AI 01-2025 — 12-2025

Università degli Studi di Padova

Padova

- Designed multimodal architectures grounding numerical concepts from visual + symbolic inputs; built robust benchmarks vs visual-only baselines.
- Implemented end-to-end data pipelines and reproducible training/evaluation on HPC; delivered systematic ablations and error analyses.
- Applied XAI to probe numerical representations; produced visualizations and interpretability summaries consumable by non-experts.
- Collaborated across research teams; documented findings and prepared materials for publications and demos.

Research Fellow - Computational Neuroscientist

03-2024 — 12-2024

Università degli Studi di Padova

Padova

- Built cognitive models (deep learning Bayesian observers) for uncertainty in time/visual perception; implemented and fitted DDM variants to behavioral datasets.
- Modeled uncertainty in visual and time perception via Bayesian observer models.
- Decision Making: implemented/fitted Evidence Accumulation models (e.g., DDM variants) on behavioural data.
- End-to-end behavioural/cognitive data analysis (experimental design, preprocessing, model fitting, validation).
- Led the analysis pipeline: experimental design, preprocessing, model fitting, validation, and reporting; co-authored a preprint.

Artificial Intelligence Engineer

10-2022 — 01-2024

ParadoxCat GmbH

Munich

- Owned radar-based hand-gesture recognition RD, achieving 92% accuracy in real-world conditions on 12 different gestures and reducing calibration time by 60% through meta-learning approaches.
- Prototyped meta-learning (MAML-style) and self-supervised (masked AE) approaches for few-shot/generalization; improved robustness under distribution shift.
- Built multimodal interior-sensing pipelines (radar + camera); optimized real-time inference on embedded hardware; delivered internal benchmarks and reusable codebase.

Deep Learning Engineer Intern

02-2022 — 07-2022

SmartEye AB

Gothenburg

- Delivered an image-to-image translation system (RGB→NIR) using state-of-the-art GANs with an encoder-decoder baseline; executed literature review, experiments, and reporting.
- Containerized training/eval with Docker; contributed C++/PyTorch utilities and internal tooling.

PROJECTS

Praenova



Co-Founder and CTO -

- B2B SaaS for AI-powered demand forecasting targeting mid-market retail, fashion, and FMCG companies. Designed the full ML architecture: multimodal signal fusion, forecasting pipelines, evaluation workflows, and explainability layer for business stakeholders. Built to reduce overstock/stockout risk with measurable supply chain impact.

EDUCATION

Master of Science in Data Science, *Università degli Studi di Padova*

Graduation in December 2023

Bachelor of Science in Scienze Psicologiche Cognitive e Psicobiologiche (L1), *Università degli Studi di Padova*

Graduation in

September 2020

PUBLICATIONS

Visalli, A., Calistrone, F. M., Calderan, M., Donnarumma, F., Zorzi, M., Ambrosini, E. (2024). A computational framework for integrating Predictive processes with evidence Accumulation Models (PAM).

Link: <https://arxiv.org/abs/2411.13203>